

Mixed-Up Cricket Bat Challenge!

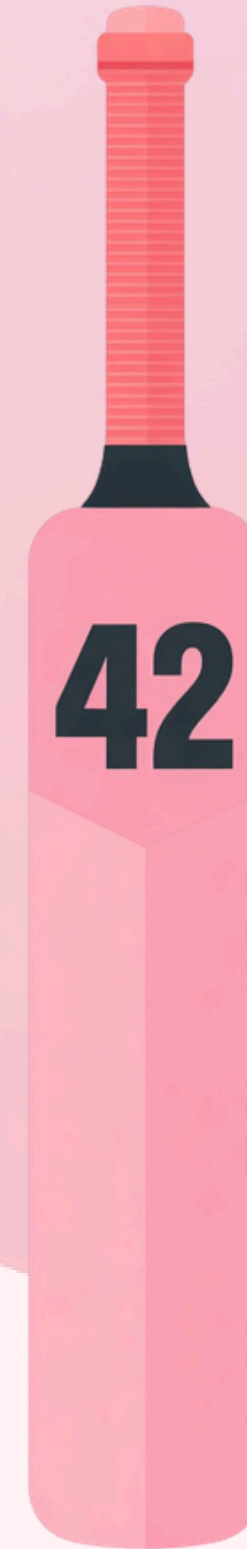
Let's find the right order, step by step!



42

18

7



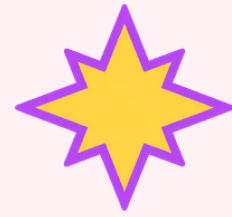
Let's Trace Pass 1 Together!

[42, 18, 7, 99, 45] → compare 42 & 18 → swap! → [18, 42, 7, 99, 45]

compare 42 & 7 → swap! → [18, 7, 42, 99, 45]

compare 42 & 99 → no swap → [18, 7, 42, 99, 45]
compare 99 & 45 → swap! → [18, 7, 42, 45, 99] ✓





Why It's Called Bubble Sort & The Temp Trick



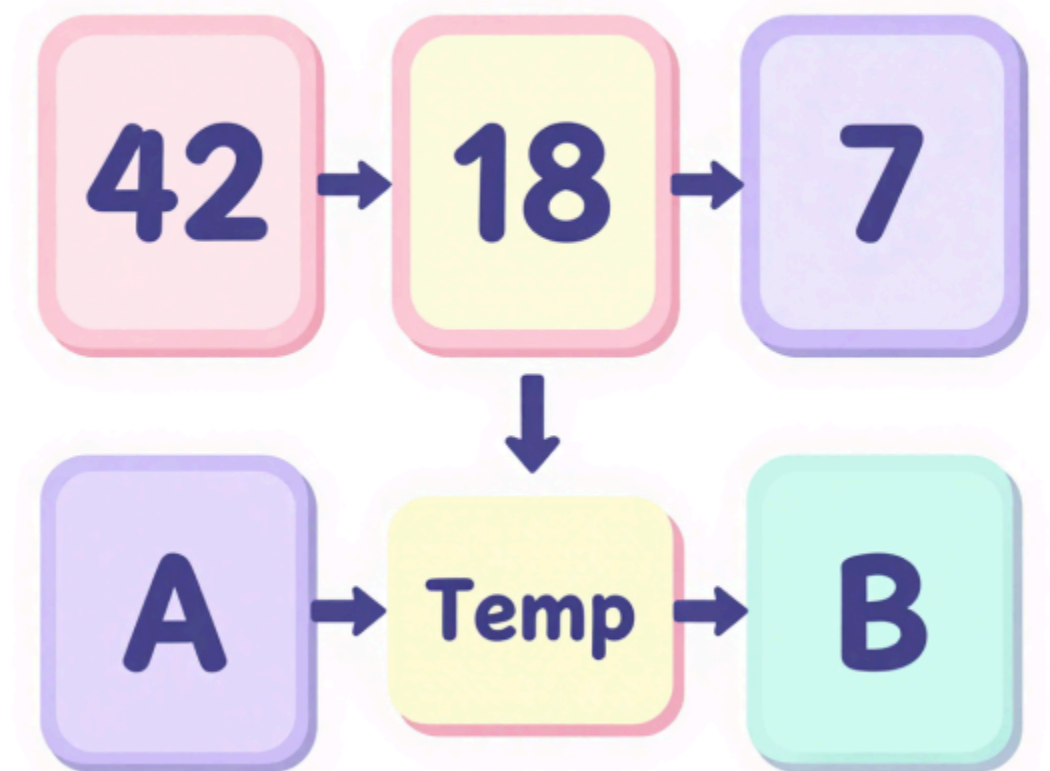
- Bigger numbers 'bubble up' like bubbles rising in nimbu pani!
- Use a 'temp glass' to hold one drink while swapping Maaza and Milk.
- Temp keeps the value safe so nothing is lost!

Bubble Sort in Blocks: Step-by-Step

 Create list called 'Scores' with your numbers

 Repeat outer loop (length - 1) times

 Inner loop: compare adjacent items
If $\text{Item}(i) > \text{Item}(i+1) \rightarrow$ use Temp to swap! 





Spot the Bug!



What happens if we forget the Temp variable when swapping?

A) The numbers stay the same 🤔

B) One number gets lost! ❌

C) The swap works perfectly ✅

D) The list gets longer 😅

Think & discuss with your partner!

❌ Without Temp

Start:

5	3
A	B

Set A = B ...

Result:

3	3
A	B

5 is LOST forever! 😱

✅ With Temp

Start:

5	3
A	B

Temp saves 5 first! 🧺

Temp=5	
3	5
A	B

Both numbers safe! 🎉

Where Do We See Sorting Around Us?



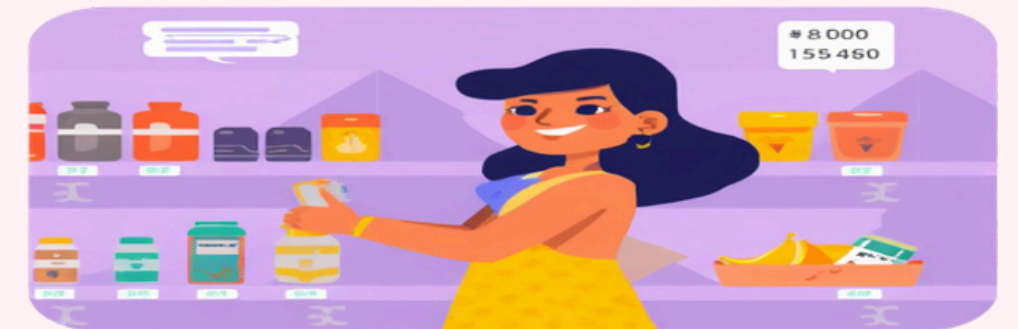
Sorting products in a shop by price or weight



Ordering train seats by ticket numbers



Arranging textbooks by weight in your school bag — heaviest at the bottom!





Great Job! Keep Sorting and Coding!
Algorithms help us solve everyday
problems step-by-step.
See you in the next session!